# Paleomagnetism of the Eocene Kamloops Group and the cratonization of Terrane I of the Canadian Cordillera

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The lower Middle Eocene (49.4 ± 2.4 Ma) Kamloops Group is exposed in the middle of the Quesnellia subterrane of Terrane I. The group consists of the siliciclastic Tranquille Beds and the overlying Dewdrop Flats plateau basalts and andesites. Detailed alternating field (AF) and thermal step demagnetization was carried out on 282 specimens from 26 flow sites and one conglomerate site, and saturation isothermal remanent magnetization (SIRM) tests were performed to examine the remanence carriers. The petrology of the gently dipping flows, the presence of antiparallel normal and reverse remanence, the conglomerate test, and the fold test all indicate that a primary remanence has been isolated. It resides in both magnetite and hematite over a broad range of AF coercivities, blocking temperatures, and domain sizes. Its mean direction of 355.0°, 73.4° ($\alpha_{95}$ = 6.9°) gives a pole position of 138.4°W, 81.4°N (dp = 11.0°, dm = 12.3°) that is statistically indistinguishable from the 50 Ma reference pole for the North American craton. This indicates that the cratonization of Terrane I was complete by the Middle Eocene after it had undergone ~1300 km of northward translation and ~45° of clockwise rotation since the mid-Cretaceous.

Le Groupe de Kamloops, d'âge correspondant au début de l'Éocène moyen (49,4 ± 2,4 Ma), affleure au milieu du subterrane Quesnellia intégré au Terrane I. Le groupe est formé des couches silicoclastiques de Tranquille et des basaltes de plateau et andésites sus-jacentes des Dewdrop Flats. Des enseignements sur les porteurs de rémanence ont été obtenus au moyen de la désaimantation par champ alternatif et par chauffage de 282 spécimens, prélevés sur 26 sites de coulées volcaniques et un site de conglomérat, et par les tests « SIRM ». La pétrologie des coulées à pendage subhorizontal, la présence de rémanence normale et rémanence inverse antiparallèles, le test de conglomérat, et le test de pli indiquent tous qu'une rémanence primaire fut isolée. La magnétite et l'hématite portent cette rémanence sur une gamme étendue de valeurs de champs alternatifs coercitifs, de températures de blocage et de grandeurs des domaines. Sa direction dominante de 355,0°, 73,4° ($\alpha_{95}$ = 6,9°) place le pôle à 138,4°O, 81,4°N (dp = 11,0°, dm = 12,3°), lequel est indiscernable, statistiquement, du pôle de référence de 50 Ma du craton de l'Amérique du Nord. Ces résultats révèlent qu'à l'Éocène moyen le craton du Terrane I était complètement formé, après avoir été déplacé de ~1300 km vers le nord et tourné de ~45° dans le sens des aiguilles d'une montre, depuis le Crétacé moyen.

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## Introduction

The Eocene Kamloops volcanics crop out extensively in the Quesnellia subterrane of Terrane I in the Cordillera of south-central British Columbia (Monger *et al.* 1982) (Fig. 1). Mesozoic rock units in Quesnellia have yielded discordant paleomagnetic pole positions that imply ~13° of northward translation and ~45° of clockwise rotation (Monger and Irving 1980; Symons 1983*a*). Conversely, the Miocene rock units have given concordant pole positions that imply neither translation nor rotation since the Miocene (Symons 1969*a*, 1969*b*). Thus the purpose of this study is to fill in the intervening time gap to determine the amount of geotectonic motion for Terrane I since the Middle Eocene. It might, in addition, shed light on the conflicting evidence of Fox and Beck (1985), who suggested about 25° of clockwise rotation for Quesnellia from the paleomagnetism of tilted Eocene volcanics in northeastern Washington but no rotation in volcanics to the immediate west.

## Geology

The Kamloops Group includes the older Tranquille Beds and younger Dewdrop Flats Formation and outcrops over an area to the west of Kamloops, British Columbia, of ~1000 km$^2$ extent (Fig. 2). The group unconformably overlies the Permian Cache Creek Group, Late Triassic Nicola Group, Early Jurassic granitic Coast Intrusions, and lower Tertiary clastic sediments.

Eocene crustal extension of this older basement led to the formation of small grabens into which the Tranquille Beds were deposited. They comprise up to 500 m of lacustrine and deltaic siliciclastic sediments including conglomerates that were derived from a volcanic terrane (Ewing 1981).

The Drewdrop Flats Formation was deposited as a flat-lying plateau consisting of up to 1400 m of basalt and andesite flows with associated breccias, tuffs, and agglomerates (Ewing 1981). Northeast of Cache Creek it is known as the Battle Creek Formation (Fig. 2). The massive flows were sampled for this study. They are fine- to medium-grained, unaltered and basaltic andesite in composition. Occasional flows are porphyritic or exhibit flow lines (Duffel and McTaggart 1952).

Ewing (1981) lists 15 K–Ar whole-rock, biotite, plagioclase, and sanidine radiometric age determinations from the Kamloops Group in the study area; they were mostly done by Mathews (1964), Hills and Baadsgaard (1967), and Church (1975). These determinations average at 49.4 ± 2.4 Ma and thus indicate a lower Middle Eocene age.

Unconformably overlying the Kamloops Group are the Miocene Cariboo plateau basalts, which form the uppermost stratigraphic rock unit in the area.

## Experimental methods and results

### *Sampling*

Three or four oriented block samples were collected from

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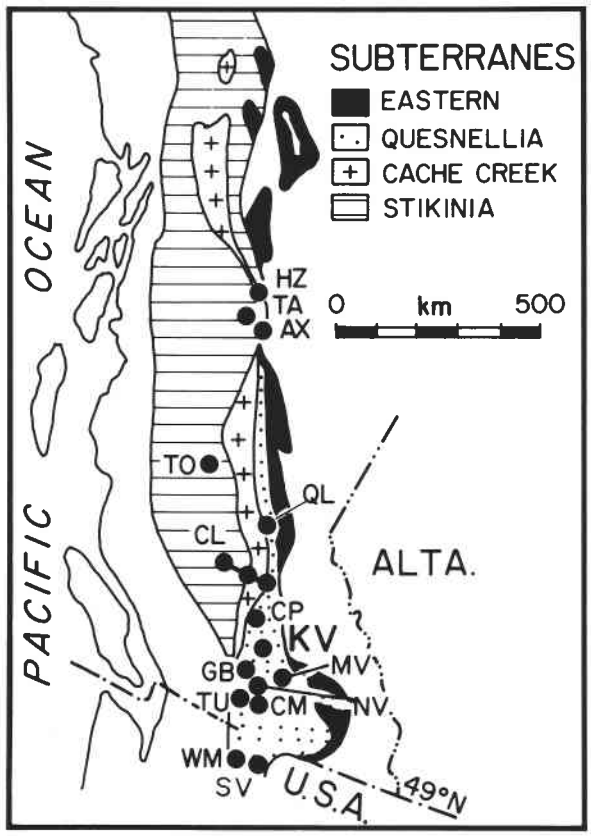



FIG. 1. Location map showing the Kamloops volcanics (KV) study area, the subterranes of Terrane I from Monger *et al.* (1982), and the locations of previous paleomagnetic studies as follows: Axelgold intrusions (AX); Cariboo lavas (CL); Copper Mountain intrusions (CM); Cariboo plugs (CP); Hazelton volcanics (HZ); Guichon batholith (GB); Marron volcanics (MV); Nicola volcanics (NV); Quesnel Lake volcaniclastics (QL); Sanpoil volcanics (SV); Takla volcanics (TA); Topley intrusions (TO); Tulameen gabbro (TU); and Winthrop – Midnight Peak sandstones (WM).

each of 26 sites (flows) in the Dewdrop Flats volcanics from rock cuts along Highway 1 and Route 156 (Fig. 2). At one additional site (22), 10 clasts were collected from the Tranquille Beds for a conglomerate test. The sites were chosen where the flow attitudes could be easily measured from flow-top breccias or interbedded agglomerates. Several oriented specimens were drilled from each block for measurement.

*Natural remanence*

The natural remanent magnetization (NRM) of all 282 specimens was measured using an automated two-axis CTF cryogenic magnetometer inside a magnetically shielded room. The NRM intensities are strong and in the $10^{-1}-10^{0}$ A · m$^{-1}$ range. Intermediate to steep northward normal polarity remanence directions were found for 19 sites (7–16, 18–21, 23–27) and relatively steep southward reversed polarity remanence directions were found for seven sites (1–6, 17).

*AF step demagnetization*

Two typical specimens from each site were alternating field (AF) demagnetized in 10 steps up to 150 mT using a Sapphire Instruments SI-4 demagnetizer. Many specimens undergo simple linear decay from the NRM direction towards the origin on a orthogonal vector decay plot, thereby indicating the presence of only one very stable remanence component (Fig. 3*a*). A majority of specimens retain low-coercivity viscous remanent magnetization (VRM) components that are removed after 10–30 mT of AF cleaning (Figs. 3*b*–3*e*). The VRM components are directed generally downwards but do not form a coherent cluster. Above ~30 mT the NRM of these specimens decays substantially and linearly towards the origin up to 120 or 150 mT, indicating that magnetite with a wide spectrum of domain sizes is a significant remanence carrier.

*Thermal step demagnetization*

A further two specimens from each site were thermally demagnetized in 11 steps up to 685°C using a Schonstedt TSD-1 demagnetizer. Again, stable normal and reverse components are shown as linear decay segments on the orthogonal vector decay plots. For most specimens the low blocking temperature VRM is removed by 200–400°C cleaning (Figs. 4*a*–4*c*), although for a few specimens cleaning above 400°C is required (Fig. 4*d*). For a majority of specimens the decay segment terminates with a significant intensity drop between 545 and 580°C as the Curie temperature of magnetite is approached, indicating that much of the remanence resides in high coercive force and blocking temperature magnetite domains (Figs. 4*a*, 4*b*). Some specimens have a significant additional intensity drop, with the same direction, between 580 and 675°C (the Néel temperature of hematite), indicating that a coexisting hematite remanence component is also present (Figs. 4*c*, 4*d*). The mean direction derived from the thermal step specimens of 353.6°, 75.5° (5.7°) (declination, inclination ($\alpha_{95}$)) is statistically the same as that derived from the AF step specimens of 352.5°, 73.4° (5.8°), meaning that the data from both sources can be validly combined.

*SIRM testing*

Seven specimens from a variety of sites with a variety of AF and thermal demagnetization characteristics were subjected to saturation isothermal remanent magnetization (SIRM) testing. Each specimen was remagnetized along one axis in a continuous field (CF) at steps of 5, 10, 20, 40, 80, 160, and 900 mT using the Sapphire Instruments SI-4 unit and a large electromagnet with 30 cm pole pieces. The very intense SIRM intensities were measured with a Schonstedt DM2000 gauss meter inside a shielded box. The acquired SIRM plot (Fig. 5*a*) shows an initial rapid increase in intensity that is indicative of magnetite, but also a consistent increase from 160 to 900 mT that is indicative of a significant percentage of hematite. When the initial part of the SIRM acquisition plot is examined more closely (Fig. 5*b*), it is evident that the remanence of specimens from sites 3, 4, and 6 is carried mostly by multidomain (MD) magnetite. The remaining specimens show an initial sharp rise from minor MD magnetite. They then track across the pseudo-single domain (PSD) and single domain (SD) magnetite curves before crossing into the hematite field, indicating that significant percentages of these remanence carriers are also present. The acquired SIRM was then AF demagnetized in seven steps up to 100 mT. The SIRM demagnetization curves (Fig. 5*c*) show similar results. The specimens from sites 3 and 6 show the dominance of MD magnetite, as well as some PSD magnetite. The remaining specimens show that PSD and SD magnetite are their main remanence carriers.

*Final demagnetization*

The remaining specimens were demagnetized in three or

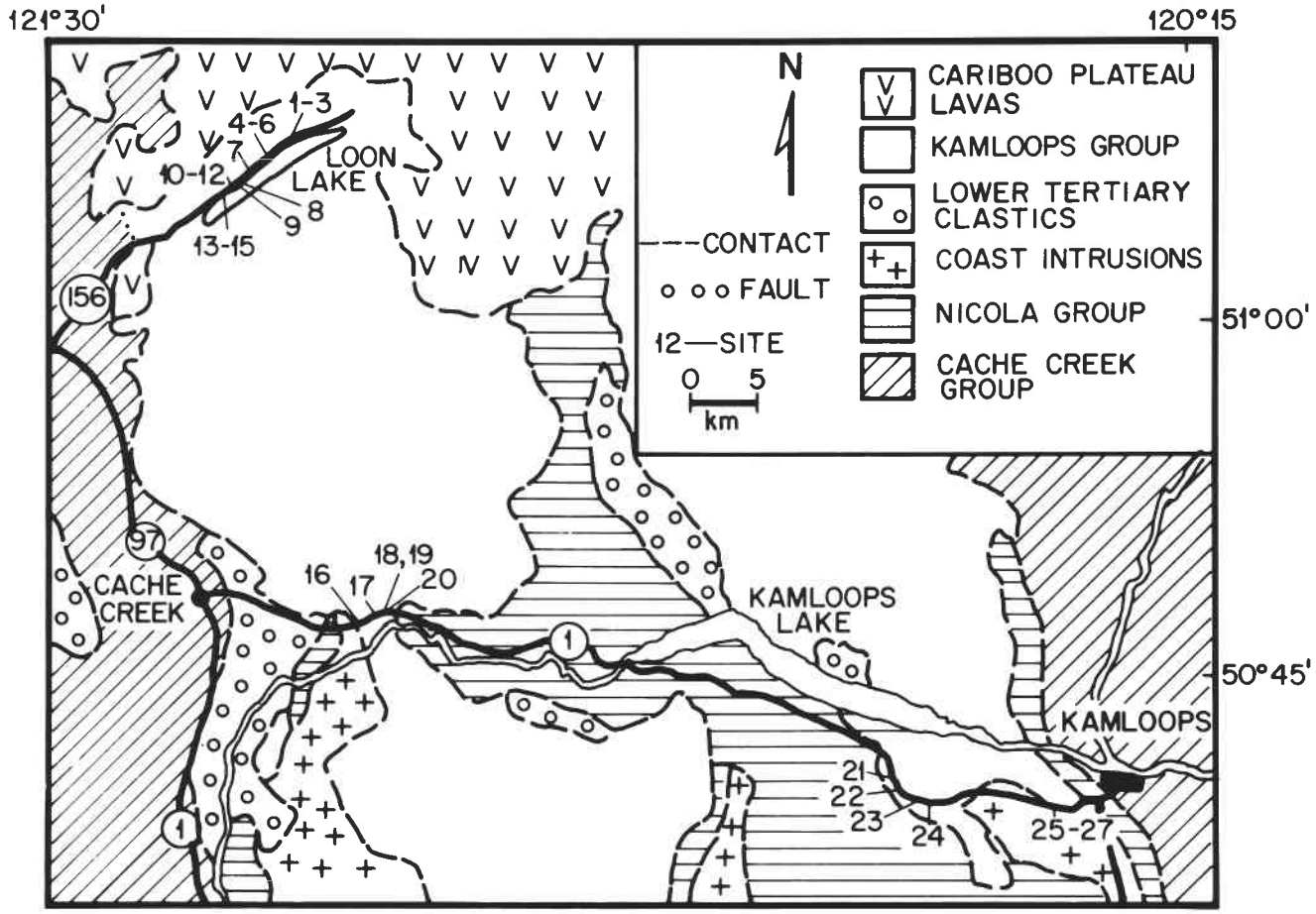



FIG. 2. Geology of the study area after Duffel and McTaggart (1952) and Cockfield (1948) with site locations.

four steps using AF intensities up to 120 or 150 mT and thermal temperatures up to 580°C in most cases. These steps were chosen to define a clear "end-point" or stable direction found by fitting two or more of the step directions, based on the demagnetization behaviour of the previously step-demagnetized specimen from the same block sample. The stable "end-point" directions for each specimen were averaged to obtain the site mean direction (Fisher 1953; Table 1). Since stable directions were observed for most specimens over a broad range of the coercivity and blocking temperature spectrums, the site means include isolation treatments ranging from NRM to 150 mT of AF cleaning and from 200 to 645°C of thermal cleaning.

Two sites (20 and 21) that failed to isolate a coherent direction and the conglomerate test site (22) were not used to calculate the unit mean direction. The remaining 24 site means after tilt correction give a unit mean direction for the Dewdrop Flats volcanics of 355.0°, 73.4° (6.9°) (Fig. 6*a*; Table 1). When specimen directions are first averaged to block mean directions and then to site mean directions, the unit mean remains statistically the same at 357.7°, 73.9° (6.7°).

*Conglomerate test*

Six of the 10 hematitic basalt clasts collected at site 22 yielded coherent specimens. All six gave very stable directions (Fig. 6*b*). When averaged, their $\alpha_{95}$ value is >90° and their resultant vector ($R$) is 2.544, which is less than the 95% confidence value of 3.85 determined by Watson (1956) for nonrandom directions. Thus the directions are random with >95% certainty and the conglomerate test is positive (Graham 1949). This means that the magnetic carriers in the clasts have not been affected by any secondary alteration or metamorphic event. Since the clasts were collected from the Tranquille Beds, it is evident that the overlying Dewdrop Flats volcanics cannot have been altered or metamorphosed and therefore retained a primary remanence.

*Reversals test*

A reversals test for remanence stability was run following McFadden and Lowes (1981). The tilt-corrected normal sites give a mean direction of 6°, 75° ($N_1$ = 17, $R_1$ = 16.0669, $k_1$ = 17.15, $\alpha_{95}$ = 8.9°); the reversed sites give 163°, −70° ($N_2$ = 7, $R_2$ = 6.8599, $k_2$ = 42.83, $\alpha_{95}$ = 9.3°); and the two populations combined give 355°, 73° ($N_c$ = 24, $R_c$ = 22.8205, $k_c$ = 19.51, $\alpha_{95}$ = 6.9°). The ratio of their precision parameters $r = k_2/k_1 = 2.50$ is just slightly greater than the $F$ statistic of 2.46 for 95% confidence. Thus the two populations have—just barely—different distributions. The probable cause is that six of the seven reversed sites are adjacent at the east end of Loon Lake (Fig. 1) and do not represent a sufficiently large population to average out secular variation, so they are abnormally well clustered. When the mean directions are compared after reversing the reversed sites, the test statistic of $r[(R_1 + R_2)^2 - R_c^2]/2[(N_1 - R_1) + r(N_2 - R_2)(R_1 + rR_2)] = 0.143$ is just less than the theoretical statistic of $(1/p)^{1/(N-2)} - 1 = 0.146$ where $p = 0.05$ for the 95%

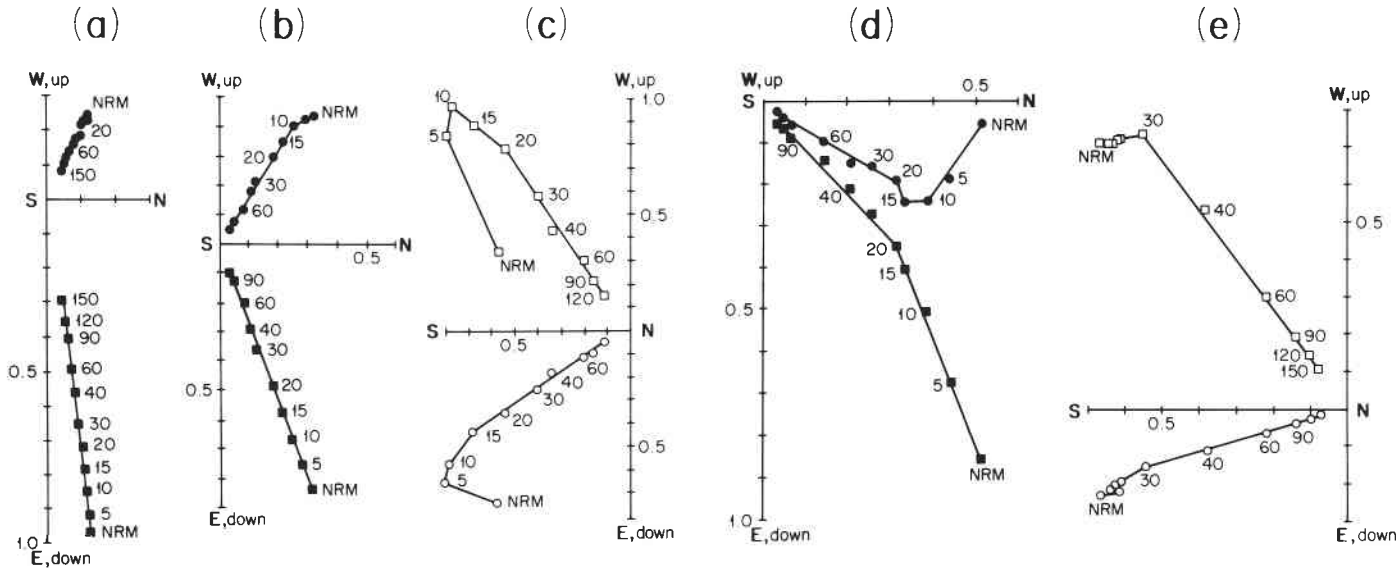



FIG. 3. Orthogonal vector decay plots for example specimens on AF step demagnetization. Circles represent vector projections on the horizontal plane (N, E, S, W) and squares, vector projections on the vertical plane (up, N, down, S); solid and open symbols denote respectively down and up directions; and the AF step intensities are in millitesla (mT). The axial values are given as a ratio of the NRM intensity. The directions are before tilt correction. The specimens come from sites 15 (*a*), 14 (*b*), 4 (*c*), 16 (*d*), and 17 (*e*).

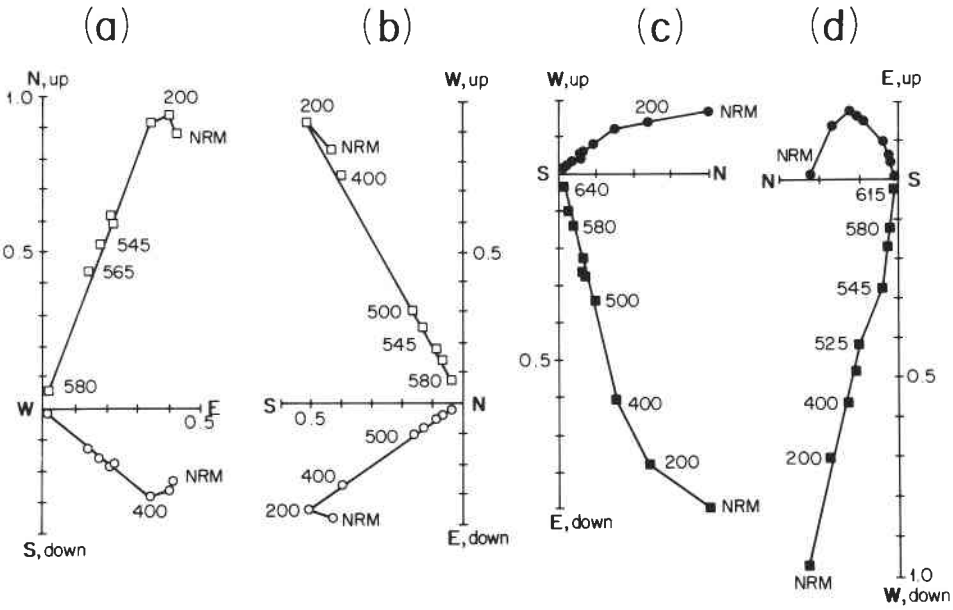



FIG. 4. Orthogonal vector decay plots for example specimens on thermal step demagnetization from sites 5 (*a*), 9 (*b*), 13 (*c*), and 8 (*d*). The plotting conventions are as given in Fig. 3 except that the thermal step temperatures are in degrees Celsius (°C).

confidence level. Therefore, the test shows that the populations are antiparallel at the 95% confidence level, meaning that it is valid to combine the populations to compute the mean and that the primary normal and reversed remanences have not been subsequently altered by a secondary event.

*Fold test*

As the dips of the Dewdrop Flats volcanics are $\leq 10°$ at all sites and average only 4°, a diagnostic paleomagnetic fold test (Graham 1949) is unlikely. Definable fold limbs are not present, so the test of McFadden and Jones (1981) is inappropriate. The site mean directions without tilt correction were averaged as before and give an uncorrected unit mean direction of 353.2°, 76.3° (8.2°). This is only ~2° from the tilt-corrected direction and is not a significant difference. However, tilt correction does decrease the $\alpha_{95}$ value from 8.2 to 6.9°, giving a corrected to uncorrected precision parameter ratio ($k_c/k_u$) of 1.40, which corresponds to an $F$ test confidence level of 87% (McElhinny 1964). In general, this test is too rigorous (McFadden and Jones 1981). Thus the fold test asserts, with at least 87% confidence, that the remanence is prefolding in origin.

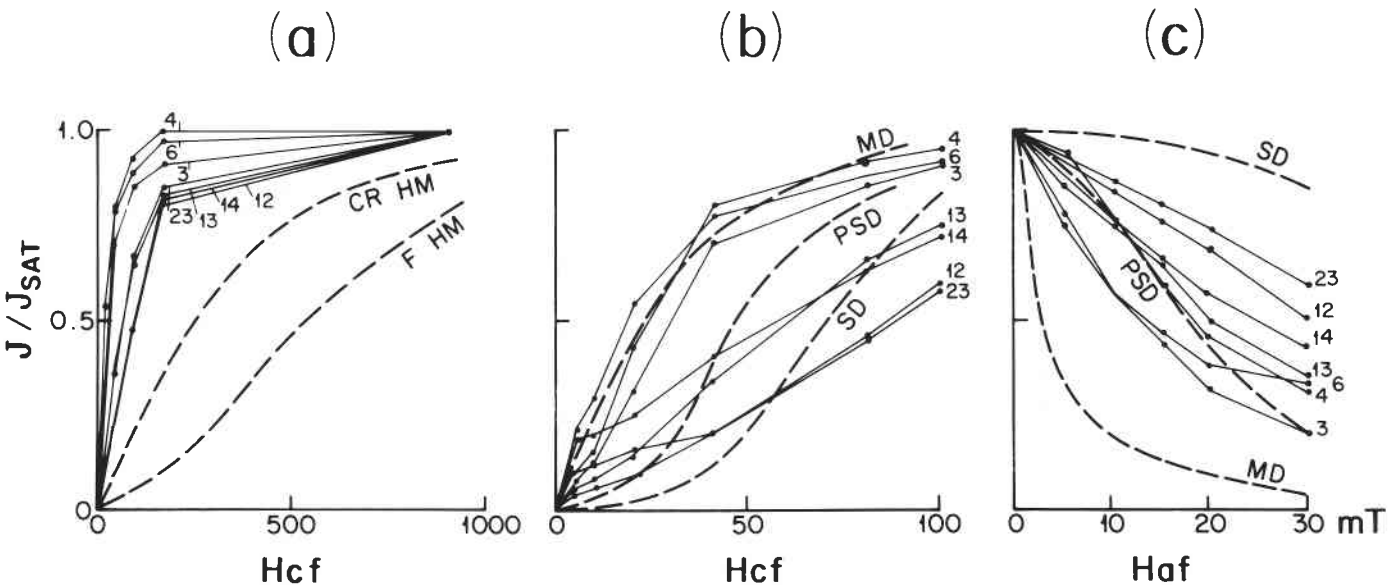



FIG. 5. Saturation isothermal remanent magnetization (SIRM) plots showing (*a*) SIRM acquisition; (*b*) initial SIRM acquisition; and (*c*) AF demagnetization of SIRM. The SIRM intensity is the measured-to-saturation (900 mT) intensity ratio ($J/J_{sat}$). The applied field (Hcf, Haf) is in millitesla (mT). The type curves for single domain (SD), pseudosingle domain (PSD), and multidomain (MD) magnetite and for coarse-grained (CR HM) and fine-grained (F HM) hematite are from Dunlop (1971, 1972, 1973, 1981). The numbers indicate the site number of the specimen.

TABLE 1. Site and unit mean directions after tilt correction

| Site No. | *N* | Treatment, °C/mT | *R* | Dec. | Inc. | $\alpha_{95}$ | *k* |
|---|---|---|---|---|---|---|---|
| 1 | 9 | 200/0−40 | 8.908 | 152.7 | −74.0 | 5.5 | 87 |
| 2 | 8 | 500/0−40 | 7.769 | 190.2 | −74.7 | 10.2 | 31 |
| 3 | 8 | 200−500/0−60 | 7.980 | 188.2 | −76.6 | 2.6 | 444 |
| 4 | 17 | 580/0−40 | 15.542 | 159.7 | −72.4 | 11.3 | 11 |
| 5 | 11 | 580−615/40−150 | 10.679 | 146.3 | −65.4 | 8.3 | 31 |
| 6 | 9 | 545−580/60 | 8.636 | 162.0 | −69.7 | 11.2 | 22 |
| 7 | 10 | 580/40 | 9.963 | 117.0 | 82.0 | 3.1 | 243 |
| 8 | 10 | 565−645/40 | 9.941 | 60.9 | 73.8 | 3.9 | 154 |
| 9 | 9 | 580−645/30−150 | 8.887 | 66.0 | 71.4 | 6.2 | 71 |
| 10 | 7 | 400−580/0−150 | 6.945 | 68.5 | 74.6 | 5.8 | 109 |
| 11 | 9 | 580/20−150 | 8.758 | 99.8 | 75.2 | 9.1 | 34 |
| 12 | 9 | 580/0−150 | 8.924 | 49.2 | 72.7 | 5.1 | 106 |
| 13 | 6 | 580/0−60 | 5.991 | 311.5 | 72.5 | 2.9 | 553 |
| 14 | 8 | 580−645/0−40 | 7.923 | 302.3 | 58.3 | 5.8 | 92 |
| 15 | 10 | 580−615/0−150 | 9.921 | 306.3 | 68.7 | 4.5 | 114 |
| 16 | 10 | 580/0−60 | 9.628 | 28.7 | 45.6 | 10.0 | 24 |
| 17 | 9 | 580/0−40 | 8.909 | 151.5 | −46.5 | 5.5 | 88 |
| 18 | 6 | 500−580/20−150 | 5.847 | 40.0 | 54.9 | 11.9 | 33 |
| 19 | 16 | 580/0−120 | 15.111 | 313.2 | 75.5 | 9.3 | 17 |
| 20 | 10 | 550−580/40−150 | 6.536 | 354.1 | 66.9 | 37.8 | 3 |
| 21 | 8 | 580/20−150 | 3.175 | 221.6 | 32.2 | 79.1 | 2 |
| 22 | 6 | 550−580/40−150 | 2.544 | 203.5 | 63.7 | >90 | 2 |
| 23 | 8 | 580−645/40 | 7.803 | 11.2 | 73.8 | 9.3 | 25 |
| 24 | 11 | 580/0−30 | 10.599 | 318.0 | 58.0 | 9.4 | 35 |
| 25 | 12 | 500−565/0−40 | 11.150 | 27.8 | 63.2 | 12.5 | 13 |
| 26 | 6 | 500/0−20 | 5.514 | 308.1 | 54.0 | 22.0 | 11 |
| 27 | 9 | 525−580/0−150 | 8.601 | 352.4 | 65.3 | 11.8 | 20 |
| Mean | 24 | | 22.820 | 355.0 | 73.4 | 6.9 | 20 |

NOTES: Sites 20 and 21 failed to retain a coherent direction and were not used in the mean calculation. Site 22 is a conglomerate test and also was not used. *N* is the number of specimens averaged for each site; *R* is the resultant vector for the summation of those specimens; Dec. and Inc. are the mean remanence vector's declination and inclination, respectively, in degrees; $\alpha_{95}$ is the radius of the cone of 95% confidence; *k* is the precision parameter of Fisher (1953).

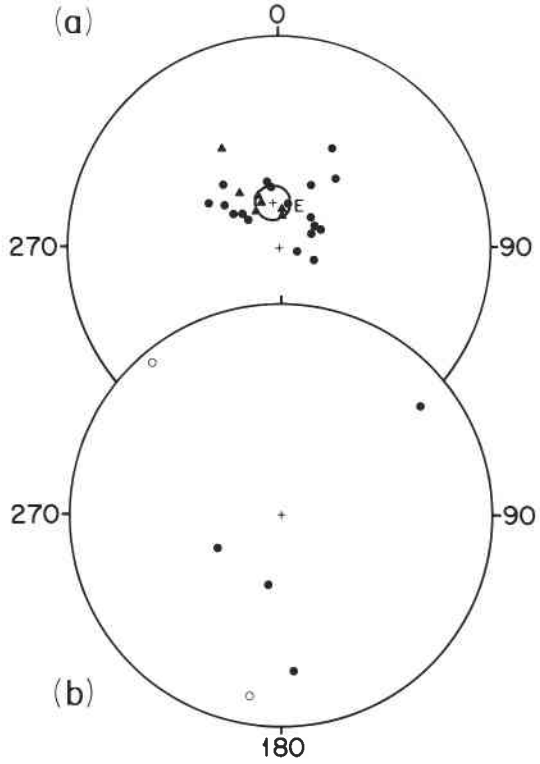



FIG. 6. Equal-area stereonet showing (*a*) the site mean directions and (*b*) the clast directions. Solid and open circles denote respectively down and up directions. Triangles are antiparallel to the "up" directions. The cross denotes the mean position with its cone of 95% confidence. *E* is the present Earth's magnetic field direction.

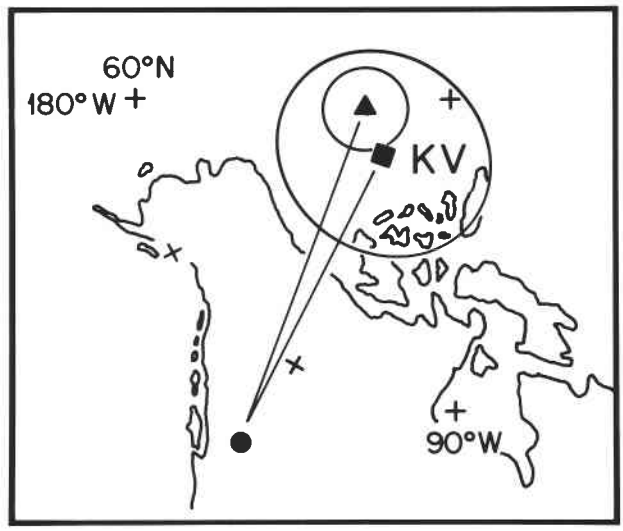



FIG. 7. Pole position for the Kamloops Group (Dewdrop Flats volcanics) (KV) shown by a square with its oval of 95% confidence. The 50 Ma reference pole for the North American craton is shown by a triangle with its circle of 95% confidence (Irving and Irving 1982, using the 30 Ma moving average). The circle is the sampling area.

## Pole position

The 49.4 Ma Dewdrop Flats volcanics show no petrologic or structural evidence of secondary alteration or metamorphism. The AF, thermal, and SIRM demagnetization data indicate that extremely stable magnetite and hematite domains are present that would require at least upper greenschist metamorphism for remagnetization to occur (Pullaiah *et al.* 1975). The presence of antiparallel normal and reverse remanence, the conglomerate test, and the fold test all indicate that a primary remanence has been isolated in the flows.

The unit mean remanence direction after tilt correction yields a paleopole at 138.4°W, 81.4°N (dp = 11.0°, dm = 12.3°) (longitude, latitude (semiaxes of the oval of 95% confidence)) (Fig. 7). This pole position is statistically indistinguishable from the 50 Ma reference pole position of 171°W, 82°N ($\alpha_{95}$ = 4°) for the North American craton (Irving and Irving 1982, using the 30 Ma smoothing window). This indicates that no statistically significant geotectonic motion has occurred between the craton and these flows since their extrusion in Middle Eocene time.

## Geotectonics

Terrane I is thought to have been assembled as an allochthonous microplate from its several subterranes in the Middle Triassic (Monger *et al.* 1982). The Late Triassic Guichon batholith and Nicola volcanics (A component), the Early Jurassic Copper Mountain intrusions and Tulameen gabbro, and the Late Cretaceous Winthrop – Midnight Peak sandstones come from south of Kamloops in the Quesnellia subterrane. All five units give similar but discordant pole positions that require Terrane I to have been translated northward by ~13° or 1300 km and rotated clockwise by ~45° relative to the craton (Symons 1974, 1983*a*, 1985; Symons and Litalien 1984; Granirer *et al.* 1986).

The paleomagnetic data from north of Kamloops in Terrane I are less certain. The Late Triassic Takla volcanics, Early Jurassic Hazelton volcanics, Late Jurassic Topley intrusions, mid-Cretaceous Axelgold anorthosite intrusion, and metamorphosed Quesnel Lake volcaniclastics, all give paleolatitudes that are consistent with a northward translation of ~1300 km for Terrane I (Monger and Irving 1980; Symons 1983*b*; Rees *et al.* 1985). A translation of this magnitude is consistent with the known displacements on right-lateral strike-slip faults on the east side of Terrane I (Gabrielse 1985, 1987). However, the Hazelton and Quesnel Lake locales in the Quesnellia subterrane and the Takla locales in the Stikinia subterrane indicate substantial but variable rotational values of −41, −103, +14, +73, +63, −43, and −62° (+ is clockwise). These locales come from folded and faulted rock units along the eastern side of Terrane I where they may well have undergone "ball bearing" type rotations of the sort proposed by Beck (1976). Conversely, the Topley intrusions come from the stable interior of Stikinia and show no rotation, suggesting that most of Terrane I to the north of Kamloops may have simply been translated northward by ~1300 km but not rotated (Symons 1971, 1983*b*). Recent paleomagnetic data from the Upper Cretaceous Carmacks volcanics in the Stikinia and Cache Creek terranes of northern British Columbia support this model (Marquis and Globerman 1987). Even if a number of these Mesozoic units have been remagnetized in Cretaceous time, as Irving *et al.* (1985) suggest, Terrane I must still have been translated ~13° northward and at least the southern part rotated ~45° clockwise since then. Also, even if the Mesozoic apparent polar wander path of Gordon *et al.* (1984) is accepted and the analysis of May and Butler (1986) is correct that the Terrane I poles are concordant in paleolatitude,

at least the southern part of Terrane I must have been rotated ~45° clockwise since then.

The concordant pole position for the Kamloops Group indicates that the cratonization of Terrane I was complete by the lower Middle Eocene. Thus the translation and rotation must have occurred between 100 and 50 Ma ago. This corresponds closely to the time interval when the Kula plate was spreading northward at ~12 cm/year along the western side of the craton to transport Terrane I northward (Coney 1978; Stone *et al.* 1982). If only rotation was involved without translation, then eastward compression from Farallon plate motion could have been responsible (Engebretson *et al.* 1985).

Bardoux and Irving (1989) have concluded that the Middle Eocene Marron volcanics in Quesnellia to the southeast of Kamloops give a concordant pole position after substantial tilt corrections, and this is consistent with the Kamloops volcanics result reported here. Both results disagree with Fox and Beck (1985), who interpreted a discordant pole from the deformed Eocene Sanpoil volcanics from further south in northeastern Washington to be recording a 25° clockwise rotation of the southern half of Quesnellia. An unrecognized local tectonic effect is the likely cause of the discordance that they observed.

Stabilization of Terrane I by the Middle Eocene is also consistent with the finding of concordant pole positions for its younger rock units. These are the Cariboo plateau lavas and Cariboo gabbro plugs that overlie and intrude the Quesnellia, Cache Creek, and Stikinia subterranes (Symons 1969*a*, 1969*b*). It also indicates that the B component pole for the Nicola volcanics, originally attributed to hematitization associated with Eocene volcanism, must have formed during an earlier alteration event (Symons 1985). It now appears that the probable event was the Early Cretaceous extrusion of the thick overlying Kingsvale volcanics, in which case the B pole reflects ~13° of translation and ~45° of clockwise rotation along with the rest of southern Quesnellia.

Relevant paleomagnetic data from the immediately outboard terranes to the west of Terrane I also support its stabilization by the Middle Eocene. The Coast Plutonic Complex yields two concordant Eocene poles. One is from the granites in the main north-northwest plutonic trend near Kitimat, British Columbia, to the west of the Topley intrusions (Symons 1977). The other is from the Hope complex to the west of the Copper Mountain intrusions (Symons 1973*a*). There are also two concordant Oligocene results from the east side of the Cascade Ranges in Washington. These are from the Chilliwack and Grotto batholiths (Beske *et al.* 1973; Beck *et al.* 1982). Thus, even though numerous studies of Eocene or younger rocks from farther west in Washington and British Columbia have shown substantial rotation (e.g., Symons 1973*b*; Magill and Cox 1981; Beck and Engebretson 1982), those abutting Terrane I appear to have been stabilized by the Middle Eocene.

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